

Gold Volume-Surge Signal: Interim Research Results

Baseline · Section A (Lookback & Normalization) · Section B (Volume-Surge Construction)
Source: executed notebook outputs (lup-2) · Close session (15:00-16:00 ET) · Target: short-horizon GLD returns

1. Research objective

This interim work tests alternative definitions of unusual gold-futures trading volume for predicting short-horizon GLD returns near the US cash close. The baseline uses a GC total-volume surge (current volume / prior 30-bar mean, expanding-normalized). Sections A and B vary lookback, normalization, and surge construction one axis at a time, then retain only constructions that improve the close-session information-coefficient (IC) screen enough to justify a full diagnostic package.

Reading note. Pearson / Spearman IC is the primary signal screen. Strategy statistics (avg bps/trade, hit rate, Sharpe, Sortino, max drawdown) are provisional diagnostics from a historical lagged 80th-percentile long-only rule with non-overlapping trades - not final production backtest results.

2. Baseline

GC total volume · 30-bar local mean · expanding normalization · Close 15:00-16:00 ET · GLD horizons 1m / 5m / 10m / 30m.

Horizon	1m	5m	10m	30m
Pearson IC	-0.0042	0.0007	0.0046	0.0166
Observations	14,607	14,607	14,607	14,607

246 trading days. Weak at short horizons; modest rise at 30m (Pearson IC ~0.0166).

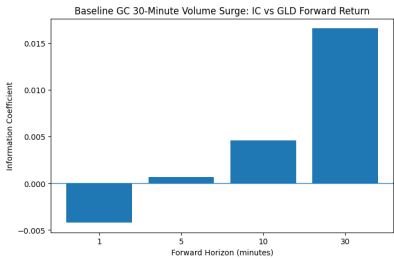


Figure: Baseline Pearson IC vs GLD forward horizon.

Pearson IC	Spearman IC	Avg bps/trade	Hit	Sharpe	Sortino	Trades	Days	Max DD (bps)
0.016617	-0.000850	0.124406	50.42%	0.122933	0.173050	476	245	-536.899

Baseline is weakly predictive at 30m (Pearson ~0.017; Spearman near zero). Provisional P&L is modest with large drawdowns relative to cumulative gains.

baseline_30_h30 | Spearman IC=-0.0009 | Pearson IC=0.0166

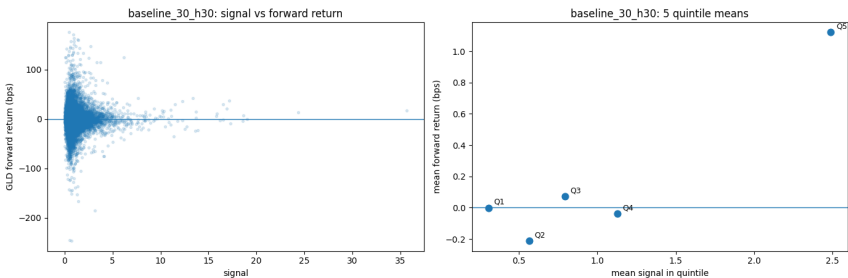


Figure: Baseline scatter and five-quintile means (Q5 is the only clear positive bucket).

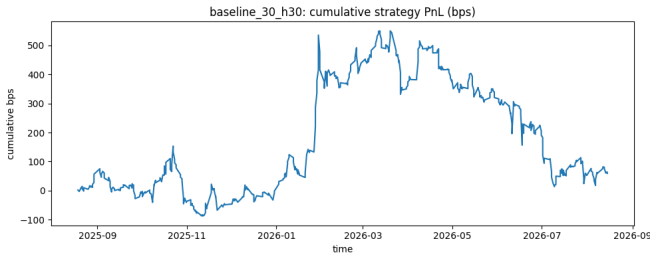


Figure: Baseline provisional cumulative P&L (bps).

3. Section A - Lookback and normalization

A.1 Lookback window

Expanding normalization and close window held fixed. Lookbacks 5 / 10 / 15 / 30 / 60 / 90 / 120 bars screened vs 1m / 5m / 10m / 30m.

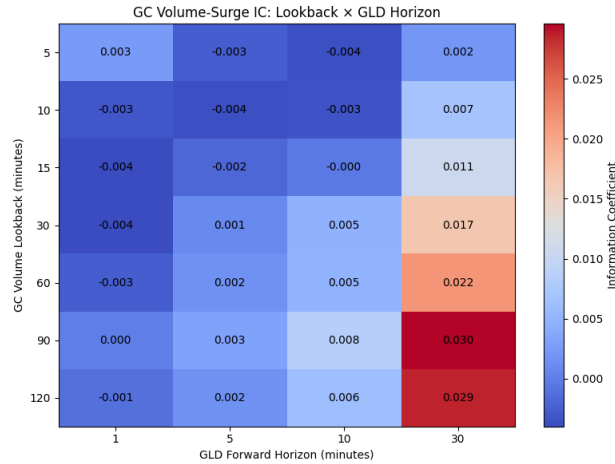


Figure: Close-window Pearson IC heatmap - lookback x GLD horizon.

Lookback	Pearson IC (close, 30m)	vs baseline
30-bar (baseline)	0.016617	-
90-bar	0.029585	+78%
120-bar	0.028562	+72%

Main result: baseline 30-bar @ 30m Pearson IC ~0.0166; 90-bar ~0.0296; 120-bar ~0.0286. Both longer lookbacks are retained.

A.1 selected-candidate diagnostics

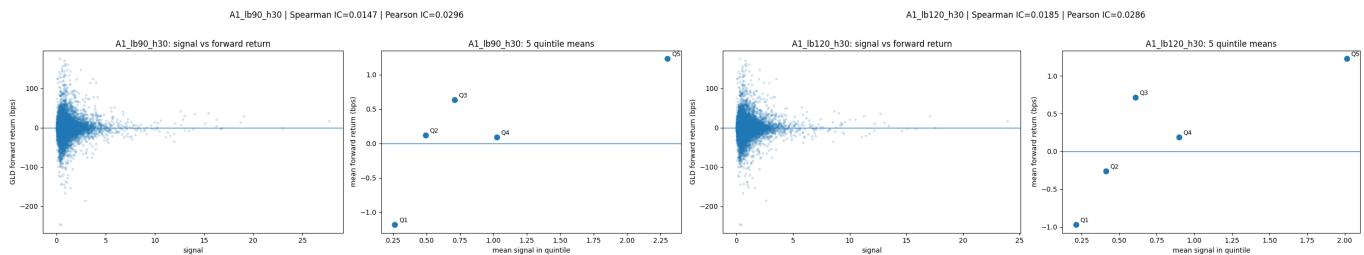


Figure: A1 90-bar (left) and 120-bar (right) scatter / quintile diagnostics @ 30m GLD.

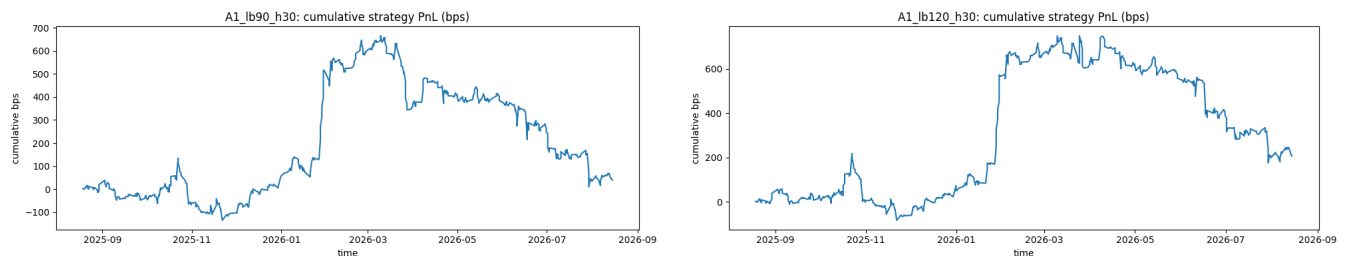


Figure: A1 provisional cumulative P&L - 90-bar (left) vs 120-bar (right). Under the provisional rule, 120m shows stronger avg bps/trade; IC and strategy rankings agree only partly.

A.2 Expanding vs rolling normalization

Lookback fixed at 30 bars. Rolling-1950 is only a marginal improvement over expanding normalization (0.0172 vs 0.0166 at 30m) and is retained as a nearby alternative, not a material upgrade. Short rolling windows are weaker.

Normalization	Pearson IC @ 30m
Expanding (baseline)	0.016617
Rolling-60	0.007401
Rolling-390	0.013870
Rolling-1950	0.017248

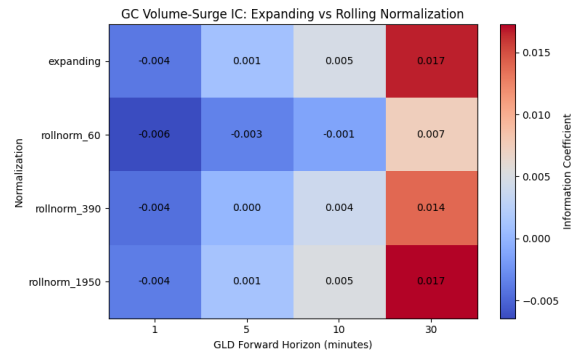


Figure (right): Expanding vs rolling IC heatmap (close window).

Section A interpretation

Longer 90/120-bar lookbacks improve close-session predictive IC at the 30-minute GLD horizon; the original 30-bar choice looks like an artifact of the initial specification. Normalization choice matters much less: expanding remains adequate.

4. Section B - Volume-surge construction

B.1-B.3 Mean / median / percentile / log

Controls fixed: 30-bar lookback, expanding normalization, close window.

Construction	@ 1m	@ 5m	@ 10m	@ 30m	Decision
Mean (baseline)	-0.0040	0.0010	0.0048	0.0166	Comparator
Median	-0.0039	0.0026	0.0050	0.0175	No new candidate
Percentile-30	0.0028	0.0006	0.0072	0.0171	No new candidate
Log-ratio	0.0001	0.0012	0.0087	0.0127	No new candidate

None produced a sufficiently meaningful improvement to create a new candidate.

B.4 Total vs uptick vs downtick

Same surge construction on GC total, uptick, and downtick volume. This axis is materially more informative than B.1-B.3.

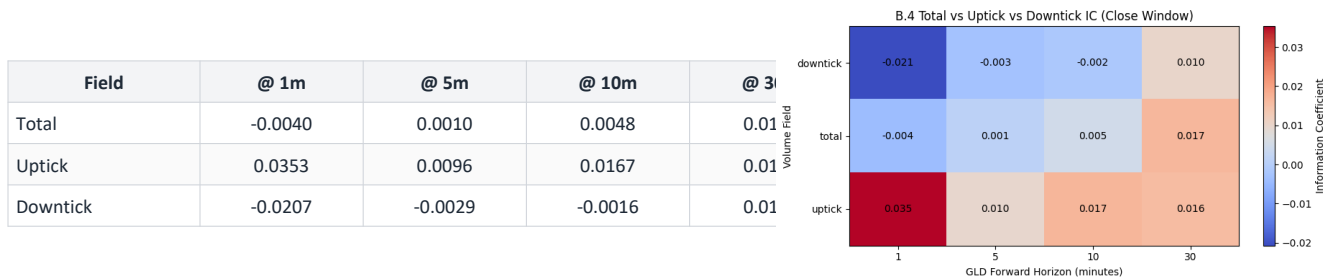


Figure (right): B.4 Pearson IC heatmap. Selected: GC uptick @ 1m - Pearson IC ~0.0353 · Spearman IC ~0.1003.

B4_uptick_h1 | Spearman IC=0.1003 | Pearson IC=0.0353

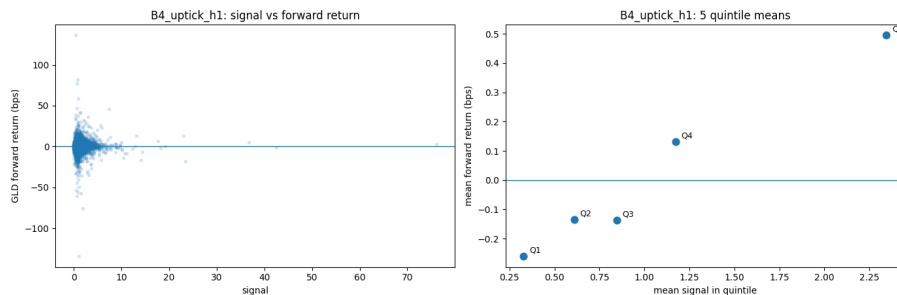


Figure: B4 uptick scatter / quintile diagnostics (monotonic Q1 to Q5).

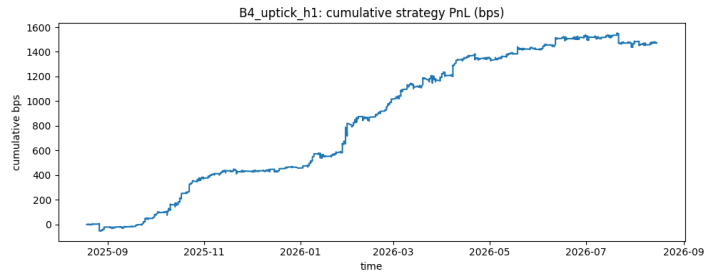


Figure: B4 uptick provisional cumulative P&L (bps). Strong on this diagnostic sample - not a production claim.

B.5 Instrument sets

Fixed 30-bar mean + expanding norm across GC, MGC, GC+MGC (oz), ETF basket (GLD+GLDM+IAU+IAUM), and GC/ETF. None improves sufficiently on the primary signal; no new candidate retained.

Construction	IC @ 30m	Decision
GC (baseline)	0.0166	Comparator
MGC	0.0140	No new candidate
GC+MGC (oz)	0.0164	No new candidate
ETF basket	0.0138	No new candidate
GC / ETF	0.0050	No new candidate

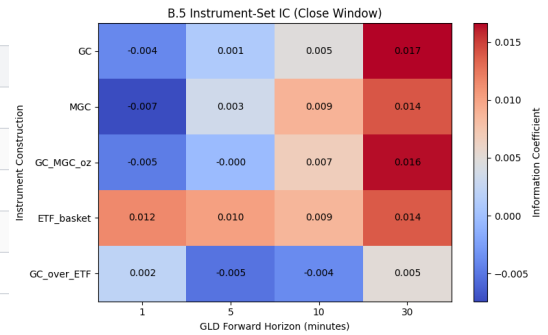


Figure (right): B.5 instrument-set Pearson IC heatmap (close window).

Section B research conclusion

Under fair one-axis evaluation, the best definition of unusual volume in this section is directional uptick intensity, not a different unsigned ratio transform. Median, percentile, log, and instrument/ETF variants are incremental or flat versus the baseline mean ratio. The Section B survivor is GC uptick-volume surge at the 1-minute GLD horizon (B4_uptick_h1), with baseline GC total-volume @ 30m remaining the common comparator.

5. Candidate comparison / current findings

Retained packages only. Strategy columns are provisional diagnostics under the notebook's lagged 80th-percentile long-only rule with non-overlapping trades - not final production backtest results. Do not treat B4 as a production-ready Sharpe-5 strategy.

Candidate	Pearson	Spearman	Avg bps	Hit	Sharpe	Sortino	Trades	Days	Max DD
Baseline 30m	0.016617	-0.000850	0.124406	50.42%	0.122933	0.173050	476	245	-536.899
A1 90m	0.029585	0.014699	0.084552	48.37%	0.082351	0.118277	459	247	-656.587
A1 120m	0.028562	0.018539	0.458565	48.67%	0.410370	0.580746	452	248	-574.655
A2 Roll-1950	0.017248	-0.001212	0.105631	49.48%	0.102123	0.147292	477	245	-543.505
B4 Uptick 1m	0.035301	0.100261	0.502709	58.38%	5.331183*	8.261305*	2,931	245	-114.860

* B4 Sharpe / Sortino are provisional diagnostic figures; Max DD in bps.

Final takeaway

The baseline GC total-volume surge is weakly predictive of GLD returns at the 30-minute close horizon. Extending the local lookback to 90 or 120 bars materially improves that close-session IC, while changing expanding versus rolling normalization adds little. Unsigned construction tweaks - median, percentile, and log-ratio - likewise add little and were not retained. Directional decomposition is much more meaningful: GC uptick volume produces the strongest Section B result (especially at the 1-minute horizon). MGC and ETF-basket additions do not improve the primary signal so far. A final optimal combined specification has not been established; retained candidates are for further joint testing.